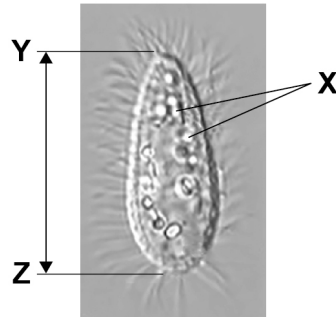


0 3

Uronema marinum is a single-celled eukaryotic organism. **Figure 3** is a photograph of *U. marinum* taken through an optical microscope.

Figure 3



0 3

. 1

Explain why it is not possible to determine the identity of the structures labelled **X** using an optical microscope.

[2 marks]

0 3

. 2

U. marinum cells ingest bacteria and digest them in the cytoplasm.

Describe the role of **one** named organelle in digesting these bacteria.

[3 marks]



0 3 . 3 Calculate the actual length of the cell shown between **Y** and **Z** in **Figure 3**.

The magnification of the image is $\times 900$

Give your answer in μm **and** to **2** significant figures.

Show your working.

[2 marks]

Answer _____ μm

0 3 . 4 In large cells of *U. marinum*, most mitochondria are found close to the cell-surface membrane. In smaller cells, the mitochondria are distributed evenly throughout the cytoplasm. Mitochondria use oxygen during aerobic respiration.

Use this information and your knowledge of surface area to volume ratios to suggest an explanation for the position of mitochondria in **large** *U. marinum* cells.

[2 marks]

Turn over for the next question

Turn over ►

